

Operating System Question Bank

Unit 4: Memory Management

1. What are the distinctions among logical, relative and physical addresses?
2. Why is the capability to relocate processes desirable? Explain in detail.
3. What are the requirements for memory management?
4. Given me, memory partitions of 100k, 500k, 200k, 300k and 600k (in-order), how would each of First-Fit, Best-Fit and Worst-Fit algorithms place processes of 212k, 417k, 112k, and 426k? Which algorithm makes the most efficient use of memory?
5. What are the steps in-handling page fault? Explain with suitable diagram.
6. Explain the address translation mechanism in paging and segmentation.
7. Write a short note on paging.
8. Explain memory management.
9. A dynamic partitioning scheme is being used, and the following is the memory configuration at a given point in time:

20M	20M	40M	60M	20M	10M	60M	40M	20M	30M	40M	40M

The shaded areas are allocated blocks; the white areas are free blocks. The next three memory requests are for 40M, 20M and 10M. Indicate the starting address for each of the three blocks using the following placement algorithms:

- i. First-fit
 - ii. Best-fit
 - iii. Worst-fit
10. Explain the concept of swapping. Also explain the swapping methods.
 11. What is contiguous memory allocation?
 12. Describe Address binding and dynamic loading.
 13. Explain the following terms in brief:
 - i. Internal fragmentation
 - ii. External fragmentation
 - iii. Compaction
 14. With the help of a diagram explain the concept of segmentation.
 15. Consider the following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- i. 0, 430
- ii. 1, 10
- iii. 2, 80
- iv. 3, 400
- v. 4, 90

16. Explain two level page table organizations for implementing virtual memory.
17. Write a short note on Buddy system.
18. For the following page reference string 5, 6, 7, 8, 5, 6, , 5, 6, 7, 8, 9, show and count the number of page faults that occur with three frames using FIFO, LRU and optimal page replacement methods.
19. Consider the following page reference string:
1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
Calculate the no. of page faults for following page replacement algorithm FIFO, Optimal and LRU.
20. What are the steps in-handling page fault? Explain with suitable diagram.
21. Explain the concept of Demand Paging with neat diagram.
22. A process references pages in the following order
5 4 2 4 6 5 3 6 2 3 2 4 4 5 2 6
Determine the number of page faults for FIFO, optimal and LRU page replacement algorithms for 3 page frames.
23. Consider the page reference string with 3 frames
A, B, C, D, E, C, D, A, F, G, H, I, G, H, I, E, D, E, D, B
Calculate the no. of page faults for following page replacement algorithms FIFO, LRU and Optimal.
24. Describe how linux implements the following aspects of memory management.
 - i. Page allocation
 - ii. Page replacement algorithm
25. Explain Belady's anomaly with suitable example.
26. What is cause of thrashing? How does the system detect thrashing? How the system can eliminate it?
27. What is allocation of frames? Explain different types of frame allocation.
28. Discuss the difference between global vs local allocation.

29. What are the various types of page replacement algorithm? Explain with an example
30. Explain the various methods to handle thrashing.